



East West University
Department of Computer Science and Engineering
Course Outline of CSE302
Spring 2026 Semester

Course: CSE302 Database Systems

Credits and Teaching Scheme

	Theory	Laboratory	Total
Credits	3	1.5	4.5
Contact Hours	3 Hours/Week for 13 Weeks + Final Exam in the 14 th Week	3 Hours/Week for 13 Weeks	6 Hours/Week for 13 Weeks + Final Exam in the 14 th Week

Prerequisite

CSE106 Discrete Mathematics

Instructor Information

Instructor: **Antu Chowdhury**
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Course Objective

This course introduces the fundamental concepts and practices of designing and implementing database systems. It also enables the student to design and perform complex query operations on relational databases. It builds the capability of optimizing the databases efficiently by applying different techniques. Knowledge of this course will be needed as prerequisite knowledge for future classes such as CSE347 Information System Analysis and Design, CSE412 Software Engineering, CSE430 Software Quality Assurance, CSE436 Multimedia Design and Development, CSE464 Advanced Database System, CSE480 Web Database Programming, and CSE488 Big Data Analytics.

Office Hours

To be announced later.

Knowledge Profile

K4: Forefront engineering specialist knowledge for practice
K5: Engineering design
K6: Engineering practice (technology)

Learning Domains

Cognitive - C3: Applying, C4: Analyzing, C5: Evaluating
Psychomotor – P2: Manipulation, P3: Precision, P4: Articulation
Affective - A2: Responding

Program Outcomes (POs)

PO1: Engineering Knowledge
PO3: Design/Development of Solutions
PO5: Modern Tool Usage
PO10: Communication

Complex Engineering Problem Solution

EP1: Depth of knowledge required
EP2: Range of conflicting requirements
EP3: Depth of analysis required

Complex Engineering Activities

EA1: Range of resources
EA2: Level of interaction

Course Outcomes (COs) with Mappings

After completion of this course, students will be able to:

CO	CO Description	PO	Learning Domains	Knowledge Profile	Complex Engineering Problem Solving/ Engineering Activities
CO1	Apply the basic concepts of relational database systems and formulate algebraic expressions for representing relations.	PO1	C3	K4	
CO2	Perform and analyze a wide range of data manipulation activities using a query language for solving database queries.	PO1	C3, C4	K4	
CO3	Create a relational data model using appropriate tools and justify the goodness of the model using other optimization techniques for database-oriented application.	PO3	C4, C5 A2	K5	
CO4	Choose and justify appropriate tools for building a relational data model and create real-life, complex database applications.	PO5	P2, P3 C3, C4, C5	K6	EP1, EP2, EP3
CO5	Demonstrate skills, present concepts, and write reports to design, build and test a real-life, complex database application.	PO10	P3, P4 A2 C3, C5	K4, K5	EA1, EA2

Course Topics, Teaching-Learning Method, and Assessment Scheme

Lecture No.	Lecture Topics	Teaching-Learning Method	CO	Mark of Cognitive Learning Levels			CO Mark	Assessment (Mark)
				C 3	C 4	C 5		
Lecture 01	Definition, Purpose of Database Management Systems, View of Data, Definition of DDL, DML, DQL, DCL, and TCL.	Lectures and discussions inside and outside the class, and other resources as necessary	CO1				3 (1 set)	Mid Term Exam (25)
Lecture 02	Database Storage Engine, Query Processor, Database and Application Architecture, Database Users, and Role of Database Administrators.	Do						
Lecture 03	Relational Data Model, Database Schema, Concept of Keys: Super keys, Candidate keys, Primary key, Alternate key, Foreign keys, Schema Diagram.	Do	CO1				7 (2 set)	
Lecture 04	Basic and Advanced Relational Algebra Operators such as Select, Project, Cartesian Product, Join, and Set operators.	Do						
Lecture 05	Assignment, Rename, Aggregate Operators, Division, and Outer Joins. Class Test 01 will be covering the Relational Model and Basic Relational Algebra Expressions [Lecture 03 – Lecture 05]	Do						
Lecture 06	Class Test 01. Introduction to SQL, Writing Basic DDL, DML, and DQL.	Do	CO2				10 (2 set)	
Lecture 07	Queries using JOIN and Natural JOIN.	Do						
Lecture 08	Null Values, Aggregate Functions and Window	Do						

Lecture No.	Lecture Topics	Teaching-Learning Method	CO	Mark of Cognitive Learning Levels			CO Mark	Assessment (Mark)
				C 3	C 4	C 5		
	Functions (RANK, DENSE_RANK and so on) Class Test 02 will be covering the Basic DDL, DML, and DQL Queries with JOIN and Natural JOIN using SQL [Lecture 06 and 07]							
Lecture 09	Class Test 02. Writing intermediate and advanced SQL using the concept of Nested Subqueries.	Do						
Lecture 10	Writing intermediate and advanced SQL using CTEs (WITH clause), Outer Joins	Do						
Lecture 11	Defining Views, concept of Materialized Views, and Authorization (DCL commands such as Grant and Revoke).	Do	CO2				5 (1 set)	
Lecture 12	Review Class							
Lecture 13	Mid Term Exam [Lecture 01 – Lecture 11]	Do						
Lecture 14	Database Design using ER Model, Types of Attributes, and Mapping Cardinalities.	Do	CO3				7 (1 set)	Final Exam (30)
Lecture 15	Practicing ER modeling, reducing ER diagrams to Relational Schemas.	Do						
Lecture 16	Extended ER features such as Generalization and Specialization, and Other Design Issues. Class Test 03 will be covering the ER modeling, including different attributes and mapping cardinalities, reducing ER diagrams to relational schemas [Lecture 13 and 14]	Do						

Lecture No.	Lecture Topics	Teaching-Learning Method	CO	Mark of Cognitive Learning Levels			CO Mark	Assessment (Mark)
				C 3	C 4	C 5		
Lecture 17	Class Test 03. Introducing Database Normalization and Functional Dependency.	Do	CO3				10 (2 set)	
Lecture 18	Types of Decomposition, Finding Closure of Attributes, and Candidate Keys.	Do						
Lecture 19	Normal Forms up to 4NF, Multivalued Dependency.	Do						
Lecture 20	Practicing Normalization based on different use cases.	Do						
Lecture 21	Introduction to Transactions, ACID properties,	Do	CO3				8 (2 set)	
Lecture 22	Serializability Theory, Determining Conflict-Serializability using Precedence Graph.	Do						
Lecture 23	Recoverable and Cascadeless Schedules, introducing phenomena such as dirty write, dirty read, non-repeatable read, phantoms, and Levels of Isolation.							
Lecture 24	Basic Concepts of Database Indexing, Ordered Indices such as Primary Index, Secondary Index, Dense Index, Sparse Index,	Do	CO3				5 (1 set)	
Lecture 25	B+ Tree Indexing and Bitmap Indices.	Do						
Lecture 26	Review class or Presentation	Do						
Lecture 27	Final Exam							

Laboratory Experiments and Assessment Scheme

Experiment	Teaching-Learning Method	CO	Mark of Cognitive Learning Levels			Mark of Psychomotor Learning Levels		CO Mark
			C 3	C 4	C 5	P 2	P 3	
LAB 01: Introduction to Oracle/MySQL Database Systems, Simple DDL and DML Queries	Lab Demonstration and Experiments, Result Analysis, and Lab Assignments	CO4						
LAB 02: Basic query pattern, ALTER TABLE queries	Do	CO4						
LAB 03: Introduction to JOIN queries	Do	CO4						
LAB 04: Introduction to group by clause, aggregate functions	Do	CO4						
LAB 05: Introduction to nested subqueries, Limiting the number of results (Oracle ROWNUM/MySQL LIMIT keyword) (Lab Mid Term will be held after this lab, addressing problem-solving using SQL)	Do	CO4						
LAB 06: Introduction to Views and Authorization (GRANT and REVOKE)	Do	CO4						
LAB 07: Designing a Database using an ER model (through an appropriate tool such as SQLDeveloper)	Do	CO4						
LAB 08: Transforming an ER model into a Relational Schema	Do	CO4						

using an appropriate tool								
LAB 09: Advanced Concepts such as Database Triggers and Term Project Frontend Development using any suitable Platform	Do	CO4						
LAB 10: Term Project Development Progress Assessment	Do							
Lab Performance (Continuous Assessment)		CO4						5
In-Lab Exercises and Lab Assignments		CO4						2
Lab Exams (Lab Mid and Lab Final)		CO4						10
Total								17

Mini Projects

Mini Project	Teaching-Learning Method	CO	Mark of Cognitive Learning Level		Mark of Psychomotor Learning Levels		Mark of Affective Learning Level	CO Mark
			C3	C5	P3	P4	A2	
Lab-based Mini Project Implementation	Group-based moderately complex design project with report writing and oral/poster presentation	CO4						3
Mini Project Reporting, Presentation, and VIVA		CO5						5
Total								8

Overall Assessment Scheme

	CO-wise Breakdown						PO-wise Breakdown				
	CO1	CO2	CO3	CO4	CO5		PO1	PO3	PO5	PO10	Total
Class Performance											5
Class Test	5	5	5	0	0		10	5	0	0	15
Mid Term Exam	10	15	0	0	0		25	0	0	0	25
Final Exam	0	0	30	0	0		0	30	0	0	30
Lab Performance and Lab Exams	0	0	0	17	0		0	0	17	0	17
Mini Project Implementation	0	0	0	3	0		0	0	3	0	3
Mini Project Report and Presentation	0	0	0	0	5		0	0	0	5	5
Total	15	20	35	20	5		35	35	20	5	100

Teaching Materials/Equipment

Text Book: Avi Silberschatz, Henry F. Korth, S. Sudarshan, *Database System Concepts*, Seventh Edition, McGraw-Hill, ISBN 0-07-352332-1

Link: <https://www.db-book.com/>

Reference Book:

- Hector Garcia-Molina, Jeffrey D. Ullman and Jennifer Widom, *Database Systems: The Complete Book*, Stanford InfoLab (2nd edition)
- Thomas Connolly, Carolyn Begg, *Database Systems: A Practical Approach to Design, Implementation and Management*, Pearson (6th edition)

Software/Tools:

- Oracle Database <https://www.oracle.com/database/>
- MySQL Database <https://www.mysql.com/>
- Other appropriate tools to design and develop a database application.

* Lecture Slides and Lab Manuals will be made available to the students during the class.

Grading System

Marks (%)	Letter Grade	Grade Point	Marks (%)	Letter Grade	Grade Point
80 and above	A+	4.00	55-59	B-	2.75
75-79	A	3.75	50-54	C+	2.50
70-74	A-	3.50	45-49	C	2.25
65-69	B+	3.25	40-44	D	2.00
60-64	B	3.00	Below 40	F	0.00

Exam Dates

Mid Term Exam Date: This will be announced later.

Final Exam Date: As per the university schedule.

Academic Code of Conduct

Academic Integrity:

Any form of cheating (physical/online), plagiarism, personification, falsification of a document, or any other form of dishonest behavior related to obtaining academic gain or the avoidance of evaluative exercises committed by a student is an academic offence under the Academic Code of Conduct and may lead to severe penalties as decided by the university's Disciplinary Committee.

For details on Student Ethics and Academic Discipline, please visit <https://www.ewubd.edu/student-rules-regulation>

Special Instructions:

- Students are expected to attend all classes and examinations
- Students will not be allowed to enter the classroom after 10 minutes of the starting time.
- For plagiarism, the grade will automatically become zero for that exam/assignment.
- Normally, there will be no make-up exam. However, in case of severe illness, death of any family member, family emergency, or humanitarian reasons, if a student misses any exam, the student **MUST** get approval for a makeup exam by written application to the Chairperson through the Course Instructor within 48 hours of the exam time. Proper supporting documents in favor of the reason for missing the exam have to be presented with the application.
- For the **final exam**, there will be **NO** makeup exam. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student misses the final exam, the student **MUST** get approval of **Incomplete Grade** by written application to the Chairperson through the Course Instructor **within 48 hours** of the final exam time. Proper supporting documents in favor of the reason for missing the final exam have to be presented with the application. **It is the responsibility of the student to arrange an Incomplete Exam within the deadline mentioned in the Academic Calendar in consultation with the Course Instructor.**
- All mobile phones **MUST** be turned to silent mode during class and exam periods.
- There is **zero tolerance for cheating** in exam. Students caught with cheat sheets in their possession, whether used or not; writing on the palm of the hand, back of calculators, chairs, or nearby walls; copying from cheat sheets or other cheat sources; copying from another examinee, etc., would be treated as cheating in the exam hall. The only penalty for cheating is **expulsion for several semesters, as decided by the Disciplinary Committee of the university.**